2. The oxygen revolution changed Earth's environment dramatically. Which of the following took advantage of the presence of free oxygen in the oceans and atmosphere?
   (A) the evolution of cellular respiration, which used oxygen to help harvest energy from organic molecules
   (B) the persistence of some animal groups in anaerobic habitats
   (C) the evolution of photosynthetic pigments that protected early algae from the corrosive effects of oxygen
   (D) the evolution of chloroplasts after early protists incorporated photosynthetic cyanobacteria

3. Which factor most likely caused animals and plants in India to differ greatly from species in nearby southeast Asia?
   (A) The species became separated by convergent evolution.
   (B) The climates of the two regions are similar.
   (C) India is in the process of separating from the rest of Asia.
   (D) India was a separate continent until 45 million years ago.

4. Large-scale, worldwide adaptive radiations have occurred in which of the following situations?
   (A) when there are no available ecological niches
   (B) after each of the big five mass extinctions
   (C) after colonization of an isolated island that contains suitable habitat and few competitor species
   (D) whenever an evolutionary innovation was needed for organisms to thrive

5. Scientists studying the origin of life have accomplished which of the following steps?
   (A) abiotic synthesis of protocells with self-replicating, catalytic RNA
   (B) formation of vesicles that use RNA as a template for DNA synthesis
   (C) formation of protocells that use DNA to direct the polymerization of amino acids
   (D) abiotic synthesis of RNA's bases (A, C, G, U)

Levels 3-4: Applying/Analyzing

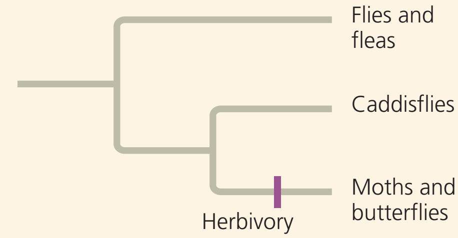
6. A genetic change that caused a certain *Hox* gene to be expressed along the tip of a vertebrate limb bud instead of farther back helped make possible the evolution of the tetrapod limb. This type of change is illustrative of
   (A) the influence of environment on development.
   (B) paedomorphosis.
   (C) a change in a developmental gene or in its regulation that altered the spatial organization of body parts.
   (D) heterochrony.

7. A swim bladder is a gas-filled sac that helps fish maintain buoyancy. The evolution of the swim bladder from the air-breathing organ (a simple lung) of an ancestral fish is an example of
   (A) exaptation.
   (B) changes in *Hox* gene expression.
   (C) paedomorphosis.
   (D) adaptive radiation.

Levels 5-6: Evaluating/Creating

8. **EVOLUTION CONNECTION** Describe how gene flow, genetic drift, and natural selection all can influence macroevolution.

9. **SCIENTIFIC INQUIRY** Herbivory (plant eating) has evolved repeatedly in insects, typically from meat-eating or detritus-feeding ancestors (detritus is dead organic matter). Moths and butterflies, for example, eat plants, whereas their "sister group" (the insect group to which they are most closely related), the caddisflies, feed on animals, fungi, or detritus. As illustrated in the following phylogenetic tree, the combined moth/butterfly and caddisfly group shares a common ancestor with flies and fleas. Like caddisflies, flies and fleas are thought to have evolved from ancestors that did not eat plants.



There are 140,000 species of moths and butterflies and 7,000 species of caddisflies. State a hypothesis about the impact of herbivory on adaptive radiations in insects. How could this hypothesis be tested?

10. **WRITE ABOUT A THEME: ORGANIZATION** You have seen many examples of how form fits function at all levels of the biological hierarchy. However, we can imagine forms that would function better than some forms actually found in nature. For example, if the wings of a bird were not formed from its forelimbs, such a hypothetical bird could fly yet also hold objects with its forelimbs. In a short essay (100–150 words), use the concept of "evolution as tinkering" to explain why there are limits to the functionality of forms in nature.

11. **SYNTHESIZE YOUR KNOWLEDGE**

In 2010, the Soufriere Hills volcano on the Caribbean island of Montserrat erupted violently, spewing huge clouds of ash and gases into the sky. Explain how the volcanic eruptions at the end of the Permian period and the formation of Pangaea, both of which occurred about 252 million years ago, set in motion events that altered evolutionary history.

For selected answers, see Appendix A.

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How has adaptive evolution occurred in stickleback fish?
Go to "What the Pelvis Can Teach Us about Evolution" at www.scienceintheclassroom.org.
Instructors: Questions can be assigned in Mastering Biology.